

USE CASE 3 — CRIME ANALYSIS

Name: Bhimireddy Bharath kumar Reddy

Email: bhimireddy.bharath99@gmail.com

Objective : Perform numerical and visual analysis to identify patterns and deeper insights from the Chicago crime data.

Tasks

1. CRIME INTENSITY BY TIME

1.1 Hourly Crime Distribution

Description : Analyze the number of reported crimes for each hour of the day to understand when crime occurs most frequently.

Code :

```
def analyze_hourly_crimes(data):  
    show_title("1. CRIME INTENSITY BY TIME")  
    show_section("1.1 HOURLY CRIME DISTRIBUTION")  
    hourly_count = (  
        data["Hour"]  
        .value_counts()  
        .sort_index()  
    )  
    print(hourly_count.to_string())
```

Output :

```
1.1 HOURLY CRIME DISTRIBUTION  
-----  
Hour  
0      16  
1      16  
2      19  
3      27  
4      23  
5      15  
6      37  
7      76  
8      67  
9      81  
10     83  
11     94  
12    108  
13     99  
14    115  
15    122  
16    129  
17    111  
18    155  
19    151  
20    153  
21    125  
22    114  
23     64
```

Insights :

- The crime records are grouped according to the hour of the day.
- This shows how crime frequency changes throughout a 24-hour period.
- The results help identify the hours with comparatively higher crime activity.

1.2 Line Plot of Crimes Per Hour of Day

Description : Create a line graph to visually represent the number of crimes recorded during each hour of the day.

Code :

```
show_section("1.2 LINE PLOT OF CRIMES PER HOUR OF DAY")

hours = hourly_count.index
counts = hourly_count.values

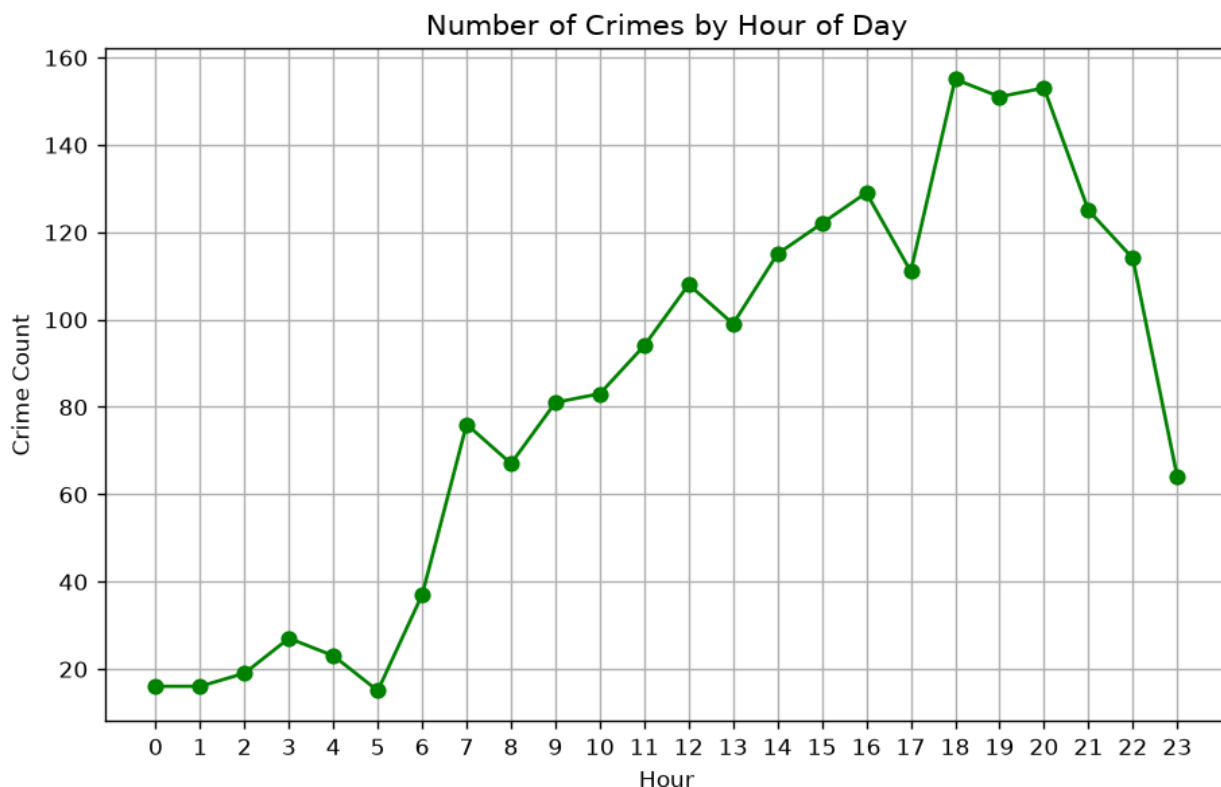
plt.figure(figsize=(9, 5))

plt.plot(
    hours,
    counts,
    marker="o",
    color="green"
)

plt.title("Number of Crimes by Hour of Day")
plt.xlabel("Hour")
plt.ylabel("Crime Count")
plt.xticks(range(24))
plt.grid()

plt.tight_layout()
plt.show()
```

Output :



Insights :

- The line graph clearly shows the variation in crime frequency throughout the day.
- Peaks in the graph represent hours with higher numbers of reported crimes.
- Lower points indicate hours with comparatively fewer crime records.
- This visualization helps identify important time-based crime patterns.

2. COMMUNITY AREA CLUSTERS USING NUMPY

2.1 Mean Crime Per Community Area

Description : Calculate the average number of crimes recorded per community area using NumPy.

Code :

```
show_section("2.1 MEAN CRIME PER COMMUNITY AREA")

average = np.mean(crime_values)

print(
    f"Mean crime per community area: {average:.2f}"
)
```

Output :

```
-----
Mean crime per community area: 25.64
```

Insights :

- The mean represents the average number of crime records per community area.
- It provides a baseline for comparing crime counts between different communities.
- Community areas with values significantly above the mean may have comparatively higher crime activity.

2.2 Extreme Outliers Using IQR Method

Description : Identify community areas with unusually high or low crime counts using the Interquartile Range (IQR) method.

Code :

```
show_section("2.2 EXTREME OUTLIERS USING IQR METHOD")|
quartiles = np.percentile(
    crime_values,
    [25, 75]
)
first_quartile = quartiles[0]
third_quartile = quartiles[1]
spread = third_quartile - first_quartile
minimum_limit = first_quartile - (1.5 * spread)
maximum_limit = third_quartile + (1.5 * spread)
unusual_areas = community_counts[
    (community_counts < minimum_limit) |
    (community_counts > maximum_limit)
]
print(f"Q1: {first_quartile:.2f}")
print(f"Q3: {third_quartile:.2f}")
print(f"IQR: {spread:.2f}")
print(f"Lower fence: {minimum_limit:.2f}")
print(f"Upper fence: {maximum_limit:.2f}")
print("\nOutlier community areas:")
print(unusual_areas.to_string())
```

Output :

```
-----  
Q1: 22.00  
Q3: 29.00  
IQR: 7.00  
Lower fence: 11.50  
Upper fence: 39.50  
  
Outlier community areas:  
community_code  
56.0      40
```

Insights :

- Q1 and Q3 represent the lower and upper quartiles of community crime counts.
- The IQR measures the spread of the middle 50% of the data.
- Values outside the lower and upper fences are identified as potential outliers.
- These outliers represent community areas with unusually low or high crime counts compared with the overall distribution.

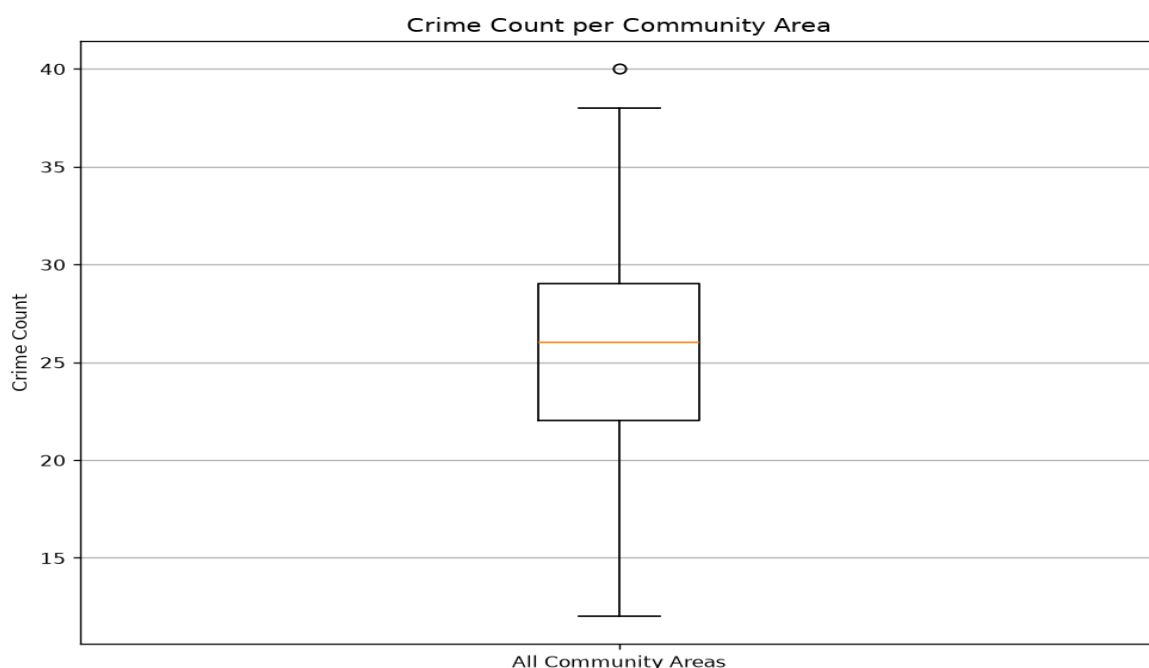
2.3 Box Plot of Crime Count Per Community Area

Description : Create a box plot to visually identify the distribution and extreme values of crime counts across community areas.

Code :

```
plt.figure(figsize=(8, 6))  
plt.boxplot(  
    crime_values,  
    vert=True  
)  
plt.title("Crime Count per Community Area")  
plt.ylabel("Crime Count")  
plt.xticks(  
    [1],  
    ["All Community Areas"]  
)  
plt.grid(axis="y")  
plt.tight_layout()  
plt.show()
```

Output :



Insights :

- The box plot shows the overall distribution of crime counts across community areas.
- The central portion represents the middle range of crime counts.
- Points outside the normal range indicate potential outlier community areas.
- This provides a quick visual understanding of variation in crime levels between communities.

3. CRIME CROSS-CORRELATION

3.1 Correlation of Numeric Features

Description : Calculate correlations between selected numerical crime-related features such as year, month, hour, beat, district, ward, community, geographical coordinates, and arrests.

Code :

```
def calculate_correlations(data):
    show_title("3. CRIME CROSS-CORRELATION")
    # Convert arrest values to integers
    data["arrest_int"] = (data["arrest"].astype(int))
    selected_features = [ "Year", "Month", "Hour",
                          "beat_num", "district_code", "ward_no",
                          "community_code", "latitude", "longitude",
                          "arrest_int"
                        ]

    show_section("3.1 CORRELATION OF NUMERIC FEATURES")
    correlation_table = (data[selected_features].corr())
    print(correlation_table.round(2).to_string())
```

Output :

3.1 CORRELATION OF NUMERIC FEATURES

```
-----
longitude  Year  Month  Hour  beat_num  district_code  ward_no  community_code  latitude
Year      1.00  0.03  0.01  -0.02      0.02      0.03      0.01      0.03
0.01      0.02
Month      0.03  1.00  0.04   0.01     -0.01     0.00     -0.00     -0.00
0.00     -0.01
Hour       0.01  0.04  1.00  -0.02     -0.02     0.03     0.01     -0.01
-0.01    -0.01
beat_num  -0.02  0.01 -0.02   1.00     0.04    -0.02    -0.02    -0.02
0.01    -0.05
district_code  0.02 -0.01 -0.02   0.04     1.00     0.01     0.01     0.00
0.01    -0.01
ward_no      0.03  0.00  0.03  -0.02     0.01     1.00     0.04     0.02
0.00      0.02
community_code  0.01 -0.00  0.01  -0.02     0.01     0.04     1.00    -0.02
-0.02      0.02
latitude      0.03 -0.00 -0.01  -0.02     0.00     0.02    -0.02     1.00
0.01      0.01
longitude      0.01  0.00 -0.01   0.01     0.01     0.00    -0.02     0.01
1.00    -0.01
arrest_int    0.02 -0.01 -0.01  -0.05    -0.01     0.02     0.02     0.01
-0.01     1.00
```

Insights :

- The correlation table shows the relationship between the selected numerical features.
- Positive correlation indicates that two variables tend to increase together.
- Negative correlation indicates that one variable tends to decrease as the other increases.
- Values close to zero indicate a weak linear relationship.

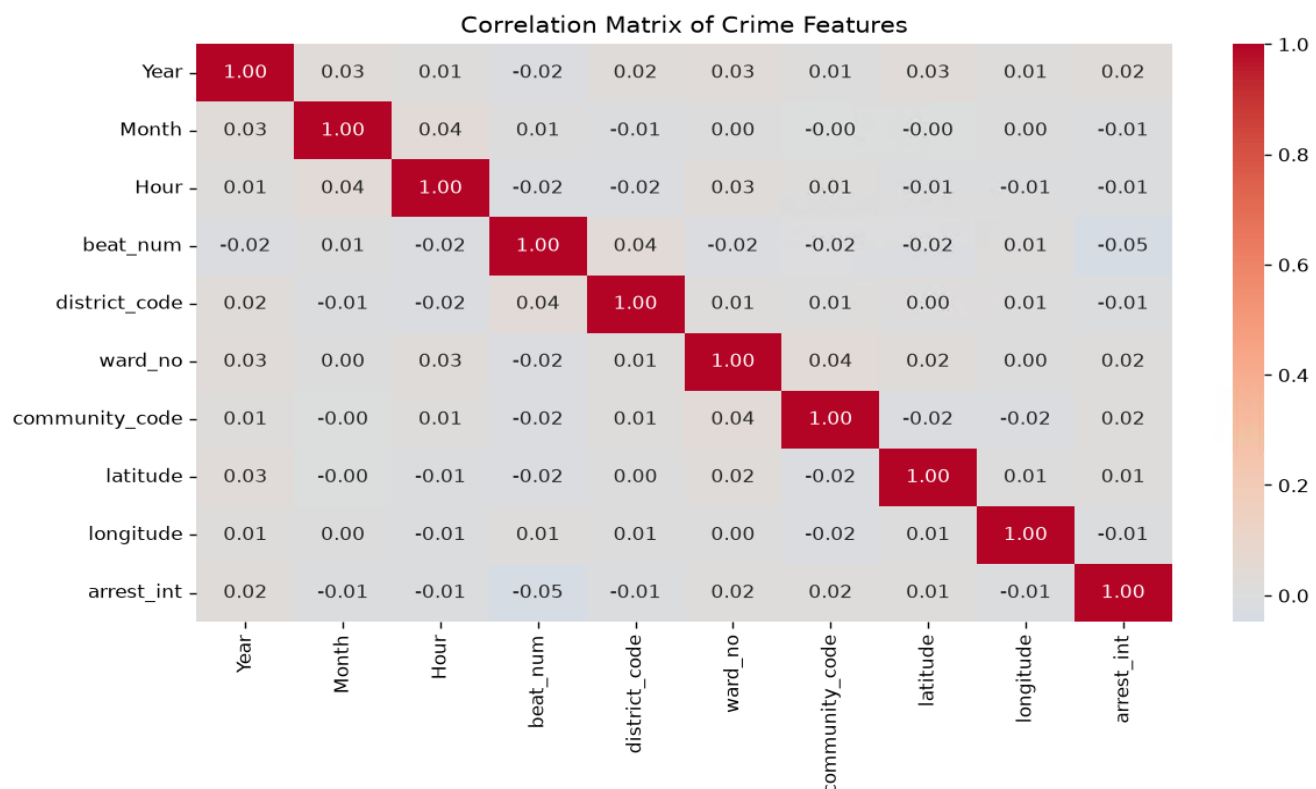
3.2 Correlation Matrix With Heatmap

Description : Visualize the correlations between numerical crime features using a heatmap.

Code :

```
show_section("3.2 CORRELATION MATRIX WITH HEATMAP")
plt.figure(figsize=(10, 8))
sns.heatmap(
    correlation_table,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0
)
plt.title("Correlation Matrix of Crime Features")
plt.tight_layout()
plt.show()
```

Output :



Insights :

- The heatmap provides a visual representation of relationships between numerical variables.
- The values inside the cells show the correlation coefficients.
- Stronger positive or negative relationships are easier to identify visually.
- This helps determine which crime-related numerical features have stronger or weaker relationships with each other.

CONCLUSION :

- Use Case 3 examines crime patterns from time, community, statistical, and correlation perspectives.
- Hourly analysis helps identify when crime occurs most frequently.
- Community-level analysis uses mean and IQR to identify unusual crime concentrations.
- Correlation analysis helps understand relationships among numerical crime features.
- The visualizations make the numerical findings easier to interpret and communicate.